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CAPSTONE PROJECT PROPOSAL

An Online Booking and Reservation Website for S-Five Inland Resort

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2025-2026

CHAPTER 1 INTRODUCTION

1.1 Background of the Study

The advancement of information and communication technology has transformed the way businesses interact with customers. In the tourism and hospitality industry, online booking and reservation systems have become essential tools for improving customer service, increasing operational efficiency, and providing convenient access to information. Resorts, hotels, and recreational facilities increasingly rely on web-based systems to manage reservations, monitor availability, and maintain customer records.

Traditionally, reservations were handled through walk-in transactions, phone calls, text messages, and manual record books. While these methods may still be used by some establishments, they often lead to delays, inaccurate records, scheduling conflicts, and customer dissatisfaction. Manual processes also require significant staff effort and are prone to human error.

The growing use of the internet has changed customer expectations. Modern customers prefer fast, reliable, and accessible services that allow them to make reservations anytime and anywhere. Online booking systems provide real-time availability checking, automated confirmation, and centralized data management. These features contribute to improved customer satisfaction and better decision-making for business owners.

S-Five Inland Resort is a recreational destination that serves guests seeking relaxation and leisure. As customer demand increases, the need for a modern reservation platform becomes more important. A web-based reservation system can streamline booking procedures, improve communication with customers, and reduce administrative workload.

This study proposes the development of an Online Booking and Reservation Website for S-Five Inland Resort. The system aims to provide a user-friendly platform where customers can create accounts, view available accommodations, submit reservations, and monitor booking status. Administrators will be able to manage reservations, customer records, and system information through a secure dashboard.

The proposed system is expected to support business growth, improve service quality, and contribute to the digital transformation of resort operations.

1.2 Statement of the Problem

The study seeks to answer the following problems:

1. How can the resort reduce delays and inaccuracies associated with manual reservation processes?
2. How can customers access real-time information regarding room and facility availability?
3. How can the resort minimize booking conflicts and duplicate reservations?
4. How can customer information be securely stored and managed?
5. How can administrators efficiently monitor and manage reservation records?
6. How can a web-based system improve customer satisfaction and operational efficiency?

1.3 Objectives of the Study

General Objective:

To design and develop an Online Booking and Reservation Website for S-Five Inland Resort.

Specific Objectives:

1. Examine the current reservation workflow of the resort.
2. Design a responsive and user-friendly interface.
3. Develop an online reservation feature with availability checking.
4. Create a database for storing customer and reservation information.
5. Develop an administrative dashboard.
6. Improve data security and system reliability.
7. Evaluate the developed system in terms of functionality, usability, and performance.

1.4 Scope and Delimitation

The study focuses on the design and development of a web-based booking and reservation system for S-Five Inland Resort. The system includes user registration, login, reservation management, availability checking, and administrative monitoring.

The system will be accessible through modern web browsers and will use a centralized database for information storage. Testing will be conducted within an academic environment and selected users.

The study excludes online payment gateway integration, mobile application development, advanced analytics, and large-scale deployment considerations.

1.5 Significance of the Study

Resort Guests: The system provides a convenient method for making reservations without requiring physical visits or phone calls.

Management: The system supports decision-making through organized records and efficient reservation monitoring.

Employees: Automation reduces repetitive tasks and minimizes errors in reservation processing.

Students: The project serves as a practical application of concepts learned in information technology and software development.

Researchers: The study may serve as a reference for future studies related to reservation systems and tourism technology.

1.6 Definition of Terms

Reservation System – A software platform used to manage bookings and reservations.

Web Application – Software accessed through a web browser over a network.

Administrator – Authorized personnel responsible for managing system data.

Database – A structured collection of information stored electronically.

Availability – The status indicating whether a room or facility can be reserved.

Client-Server Architecture – A computing model where clients request services from a server.

User Interface – The visual environment through which users interact with a system.

Authentication – The process of verifying a user's identity.

Usability – The degree to which a system is easy to use.

Reliability – The ability of a system to perform consistently and accurately.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction to the Review of Related Literature and Studies

This chapter presents literature and studies related to online booking and reservation systems, web-based information systems, tourism management, database management, and customer service technologies. The review provides a foundation for understanding the significance of the proposed Online Booking and Reservation Website for S-Five Inland Resort.

The purpose of reviewing literature and studies is to identify existing concepts, technologies, methodologies, and findings that support the development of the proposed system. It also helps determine research gaps that the present study intends to address.

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2.2 Foreign Literature

Online booking systems have become a standard feature in the hospitality industry worldwide. Researchers emphasize that digital reservation systems improve operational efficiency, customer satisfaction, and information accessibility.

Modern reservation platforms provide real-time availability checking, automated confirmations, centralized databases, and secure transaction management. These systems reduce manual workload and support better business decision-making.

The increasing adoption of cloud technologies, web applications, and mobile platforms has further enhanced reservation system capabilities. Businesses can now offer services twenty-four hours a day while maintaining accurate records and efficient communication with customers.

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Studies also reveal that user-friendly interfaces and system reliability are major factors influencing customer satisfaction. Organizations implementing reservation systems reported improved operational performance and higher customer retention rates.

Additional studies highlight the importance of data security, responsive design, and automated notifications in ensuring successful system adoption.

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2.4 Local Literature

In the Philippines, businesses increasingly adopt information systems to improve service delivery. Tourism establishments utilize reservation platforms to support digital transformation and enhance customer experiences.

Local literature emphasizes the need for affordable and user-friendly systems suitable for small and medium enterprises. Effective reservation systems contribute to organizational efficiency and improved customer communication.

The growth of internet accessibility and smartphone usage has accelerated demand for online booking services among Filipino consumers.

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2.5 Local Studies

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Several projects developed by academic institutions focused on web-based booking solutions for resorts, hotels, and tourism establishments. Findings consistently indicate that automation reduces errors and enhances service quality.

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2.6 Related Systems

Various reservation systems currently exist in the hospitality industry. These systems typically provide account registration, reservation processing, room management, availability monitoring, and administrative functions.

While many commercial systems offer advanced features, smaller establishments often require customized solutions that align with their operational requirements and budget constraints.

The proposed system aims to incorporate essential reservation functionalities while maintaining simplicity, usability, and cost-effectiveness.

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2.7 Synthesis of the Reviewed Literature and Studies

The reviewed literature and studies indicate that online reservation systems significantly improve operational efficiency, customer satisfaction, and information management. Digital platforms reduce manual errors, provide real-time access to information, and support better decision-making.

Both foreign and local studies consistently emphasize the importance of usability, reliability, and security in reservation systems. These findings support the development of an Online Booking and Reservation Website for S-Five Inland Resort.

The proposed study contributes to existing knowledge by providing a customized solution designed specifically for the operational needs of the resort while applying established principles from previous research.

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2.8 Research Gap

Although numerous reservation systems have been developed, many establishments continue to rely on manual or semi-digital procedures. Existing solutions may not fully address the specific requirements of smaller resorts.

This study seeks to bridge this gap by developing a reservation platform tailored to S-Five Inland Resort. The system focuses on usability, accessibility, reservation management, and data organization.

The research also contributes to academic knowledge by evaluating system performance and user acceptance within a real-world context.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

This study utilizes the developmental research design. Developmental research focuses on the design, development, implementation, and evaluation of systems intended to solve identified problems. The approach is appropriate because the primary objective of the study is to create an Online Booking and Reservation Website for S-Five Inland Resort.

The developmental research method allows researchers to identify user requirements, design system components, develop functional modules, test the system, and evaluate performance. The process ensures that the final product meets user expectations and operational requirements.

The study follows a systematic process beginning with requirement gathering and ending with system evaluation. Through continuous feedback and refinement, the proposed system is expected to achieve acceptable levels of functionality, reliability, usability, and efficiency.

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3.2 System Development Methodology

The Agile Development Model serves as the framework for system development. Agile emphasizes iterative development, collaboration, and continuous improvement. The methodology allows developers to adapt to changing requirements and improve the system throughout development.

Planning Phase:

The researchers gather requirements through interviews, surveys, and observation. Functional and non-functional requirements are identified and documented.

System Design Phase:

The design phase includes database design, interface design, process modeling, and system architecture planning. Diagrams and prototypes are prepared to guide development.

Development Phase:

Coding activities are performed using selected technologies. Individual modules are created and integrated into the complete system.

Testing Phase:

Testing ensures that all system components function properly. Bugs and errors are identified and

corrected.

Implementation Phase:

The completed system is deployed in a controlled environment for evaluation and user feedback.

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Coding activities are performed using selected technologies. Individual modules are created and integrated into the complete system.

Testing Phase:

Testing ensures that all system components function properly. Bugs and errors are identified and corrected.

Implementation Phase:

The completed system is deployed in a controlled environment for evaluation and user feedback.

The Agile Development Model serves as the framework for system development. Agile emphasizes iterative development, collaboration, and continuous improvement. The methodology allows developers to adapt to changing requirements and improve the system throughout development.

Planning Phase:

The researchers gather requirements through interviews, surveys, and observation. Functional and non-functional requirements are identified and documented.

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The proposed system follows a client-server architecture. Users access the application through web browsers connected to the internet or local network. Requests are transmitted to the server where application logic processes user actions.

The database server stores user accounts, reservation records, room information, and administrative data. Communication between components ensures accurate data retrieval and processing.

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3.4 Tools and Technologies Used

Frontend Technologies:

HTML, CSS, and JavaScript are used to develop the graphical user interface. These technologies

support responsive and interactive web pages.

Backend Technologies:

PHP is utilized for server-side processing. It manages business logic, user authentication, reservation processing, and database communication.

Database:

MySQL serves as the database management system. It stores customer information, reservation details, room records, and administrative data.

Development Environment:

Visual Studio Code provides coding, debugging, and project management capabilities.

Additional Tools:

Web browsers, diagramming software, and testing tools support development activities.

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3.5 Data Gathering Techniques

Interview:

Interviews are conducted with resort personnel and potential users to identify requirements and operational challenges.

Survey:

Questionnaires are distributed to gather user preferences, expectations, and feedback regarding online reservation systems.

Observation:

Researchers observe current reservation procedures to understand workflow and identify inefficiencies.

Document Review:

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3.6 System Requirements

Hardware Requirements:

Computer systems with adequate processing power, memory, storage, and network connectivity.

Software Requirements:

Operating systems, web browsers, database servers, and development tools required for system operation.

Network Requirements:

Reliable internet or local area network connectivity to support user access and server communication.

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3.7 Database Design

The database consists of multiple interconnected tables.

User Table:

Stores customer information including account credentials and contact details.

Reservation Table:

Stores booking details, reservation status, and scheduling information.

Room Table:

Contains room descriptions, pricing, and availability status.

Admin Table:

Stores administrative account information and access privileges.

Relationships between tables ensure data consistency and support efficient retrieval.

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3.8 Testing Procedures

Unit Testing:

Individual modules are tested separately to verify correct functionality.

Integration Testing:

Combined modules are evaluated to ensure proper interaction.

System Testing:

The complete application is tested against requirements.

User Acceptance Testing:

Actual users evaluate the system and provide feedback regarding usability and functionality.

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3.9 Evaluation Criteria

The developed system is evaluated according to ISO/IEC 25010 quality standards.

Functionality:

Measures whether the system performs required tasks accurately.

Usability:

Determines ease of use and user satisfaction.

Reliability:

Measures consistent performance under normal operating conditions.

Efficiency:

Evaluates resource utilization and response times.

Security:

Assesses protection of sensitive information and access control mechanisms.

Maintainability:

Measures ease of modification and future enhancement.

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3.10 Supporting Diagrams

Data Flow Diagram (DFD):

Illustrates the movement of information between users, processes, and database components.

Entity Relationship Diagram (ERD):

Shows relationships among database entities including users, reservations, rooms, and administrators.

Use Case Diagram:

Identifies system actors and interactions.

Flowcharts:

Represent procedural workflows and system processes.

These diagrams provide visual representation of system structure and operation.

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Chapter 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The **Online Booking and Reservation Website for S-Five Inland Resort** is a web-based application that allows customers to view resort information, check cottage and room availability, make online reservations, and receive booking confirmations. The system also enables the resort administrator to manage bookings, customer information, resort facilities, schedules, payments, and reports. By replacing the manual reservation process with an automated online platform, the system improves efficiency, minimizes booking conflicts, and enhances customer satisfaction.

Objectives of the developed system:

- Automate the booking and reservation process.
- Allow customers to reserve cottages and rooms online.
- Display real-time availability of facilities.
- Manage customer records and reservation history.
- Generate booking and sales reports.
- Improve customer convenience and business operations.

4.3 System Development Methodology

Use the **Agile Software Development Methodology** with the following phases:

- Requirements Gathering
- System Planning
- System Design
- Software Development

- Testing
- Deployment
- Maintenance

4.4 System Architecture

Use a **Three-Tier Architecture** consisting of:

- **Presentation Layer** – Customer and Admin interface.
- **Application Layer** – Reservation processing, authentication, booking management, payment processing, and report generation.
- **Data Layer** – MySQL database storing users, reservations, cottages, rooms, payments, schedules, and reports.

4.5 Hardware and Software Requirements

Hardware

- Intel Core i3/i5 Processor
- 8GB RAM
- 256GB SSD
- Stable Internet Connection

Software

- Windows 11
- Visual Studio Code
- PHP
- MySQL
- XAMPP
- HTML
- CSS
- JavaScript
- Google Chrome

4.6 Database Design

Suggested database tables:

- Users
- Customers
- Cottages
- Rooms
- Reservations
- Payments
- Reports
- Admin

Chapter 5

5.2 Summary of Findings

The developed website successfully:

- Automated the reservation process.
- Reduced manual booking errors.
- Improved reservation management.
- Allowed customers to reserve online anytime.
- Generated reports for the resort administrator.

5.3 Conclusion

The **Online Booking and Reservation Website for S-Five Inland Resort** successfully achieved the objectives of providing a faster, more efficient, and reliable reservation system. The website simplifies booking management, improves customer convenience, and enhances operational efficiency by replacing manual reservation methods with a secure and centralized online platform.

5.4 Recommendations

- Add online payment integration (GCash, Maya, Credit/Debit Card).
- Develop a mobile application.
- Integrate SMS and email booking notifications.
- Implement QR-code check-in.
- Add customer reviews and ratings.
- Enhance security using two-factor authentication.
- Include analytics and occupancy reports.